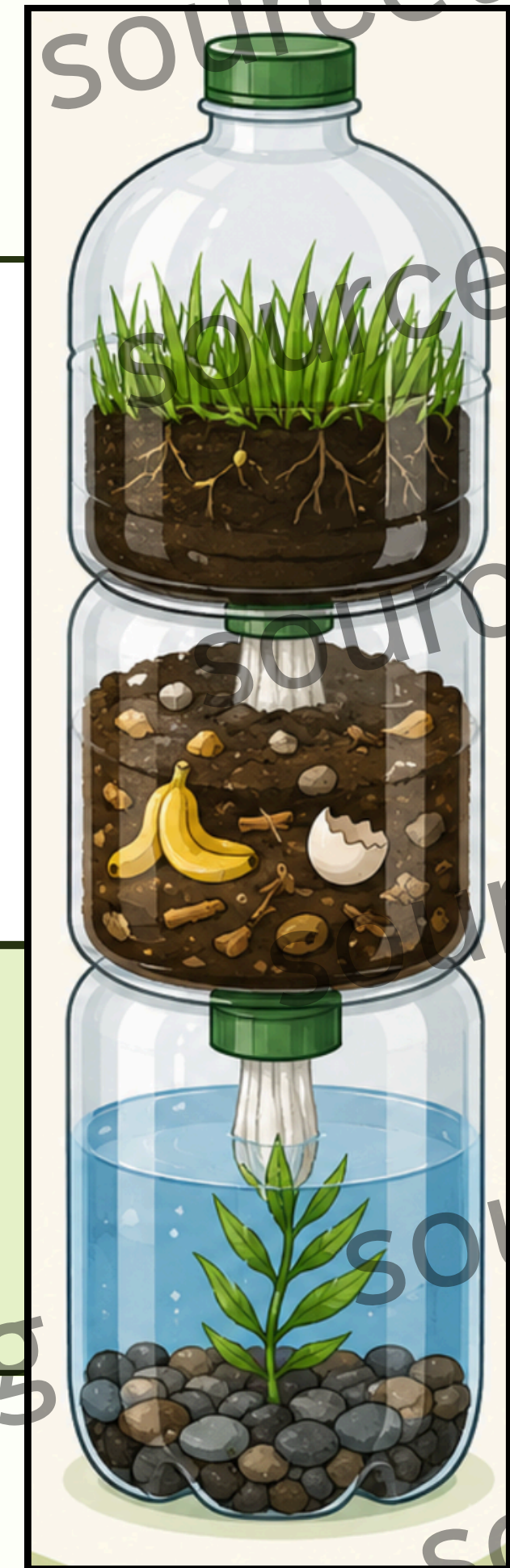


SOURCE TO SOUND

CREATING AN ECOCOLUMN

Learn how water, nutrients, and organisms move through connected ecosystems by building and testing a living model in a bottle



WHAT IS AN ECOCOLUMN?

An EcoColumn is a model built out of 3 two-liter bottles stacked on top of each other, each one representing a different part of an ecosystem.

TERRESTRIAL

- Potting soil
- Mixed grass seeds

DECOMPOSITION

- Potting soil
- Banana peels
- Eggshells

AQUATIC

- Gravel
- Aquatic plant
- Water

Water flows down through each layer, carrying nutrients and bacteria from the soil and decomposition chambers into the aquatic chamber below. Watch how the ecosystems change over time!

1

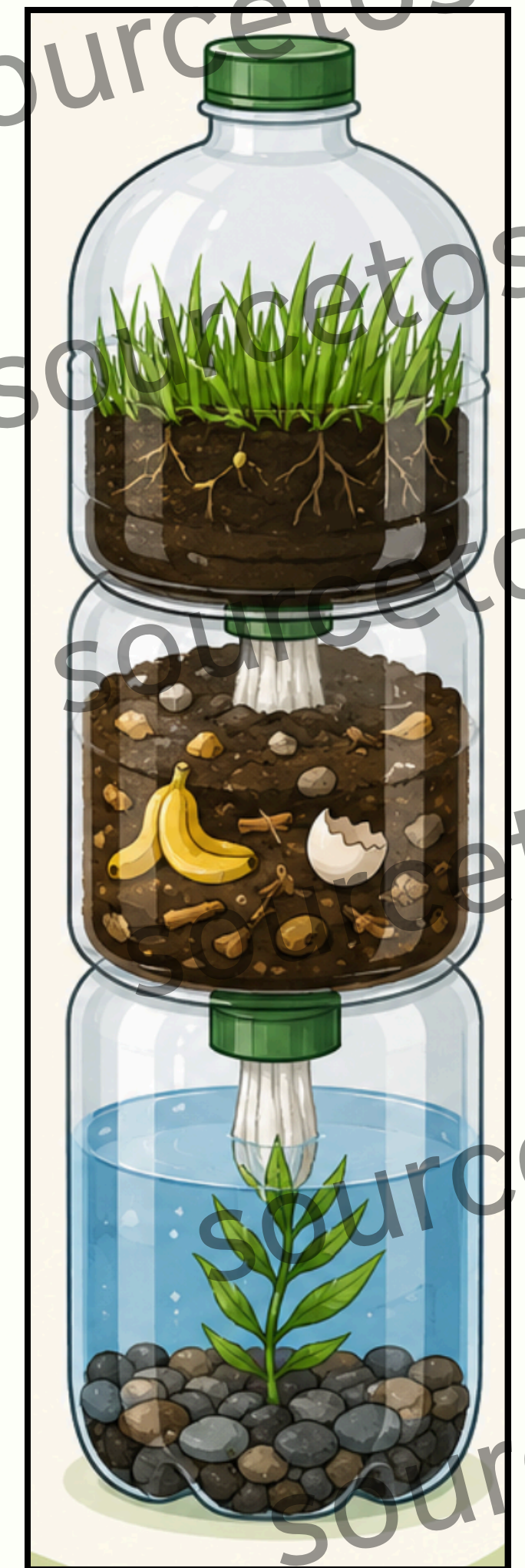
Terrestrial

2

Decomposition

3

Aquatic



BUILDING YOUR HYPOTHESIS

A hypothesis is a testable prediction about what will happen in your EcoColumn.

FOLLOW THIS FORMAT!

If [condition], then [outcome], because [reasoning]

EXAMPLE: If the aquatic chamber's nitrate levels decrease due to nutrient cycling, then the dissolved oxygen (DO) levels will increase above 5 mg/L, because less nutrient pollution will lead to less algae consuming oxygen.



Write down 3-4 different hypotheses predicting what happens to the EcoColumn over time!

TIPS



Use numbers when possible



Be specific



Connect cause to effect



Think about all three chambers

MATERIALS LIST

3 two-liter soda bottles

Clear, labels removed, rinsed,
caps with drilled holes

Gravel

**1 piece of
cheese cloth**

Compost materials

Seeds, banana peels,
eggshells

Potting soil

**Clear packing tape,
Utility knife, scissors**

Treated water

**Mixed grass
seeds**

1 Aquatic plant
Elodea is preferred

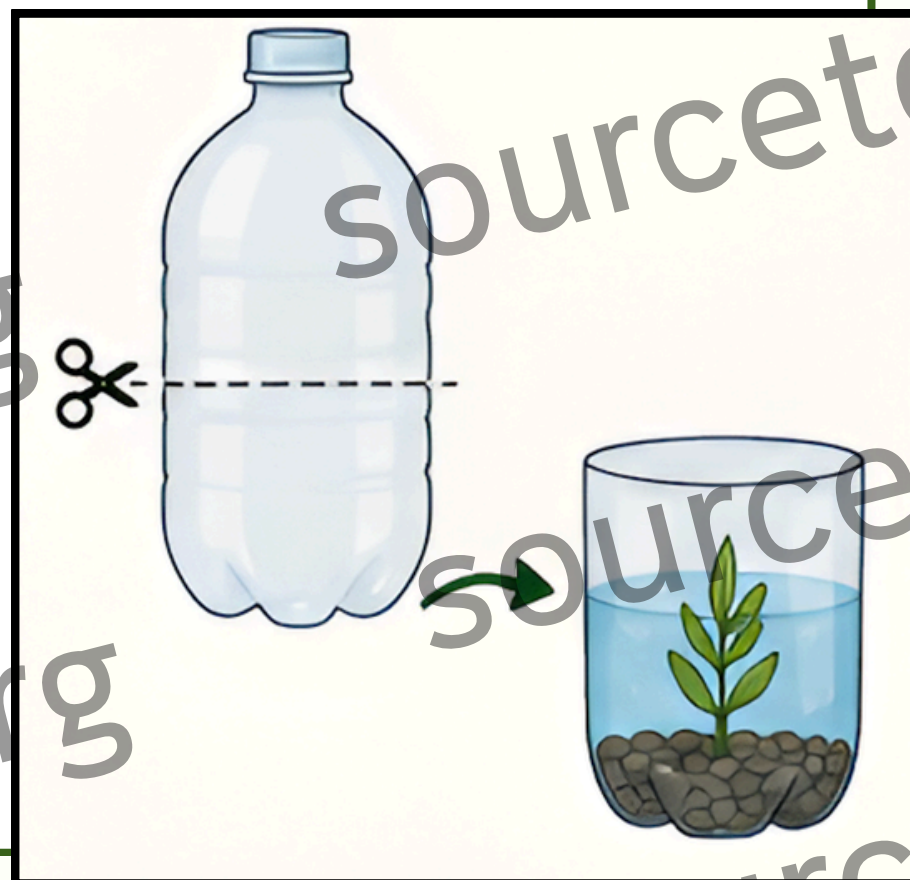


Get Creative! Think about what additional materials you would want to add to your EcoColumn.

BUILDING THE ECOCOLUMN

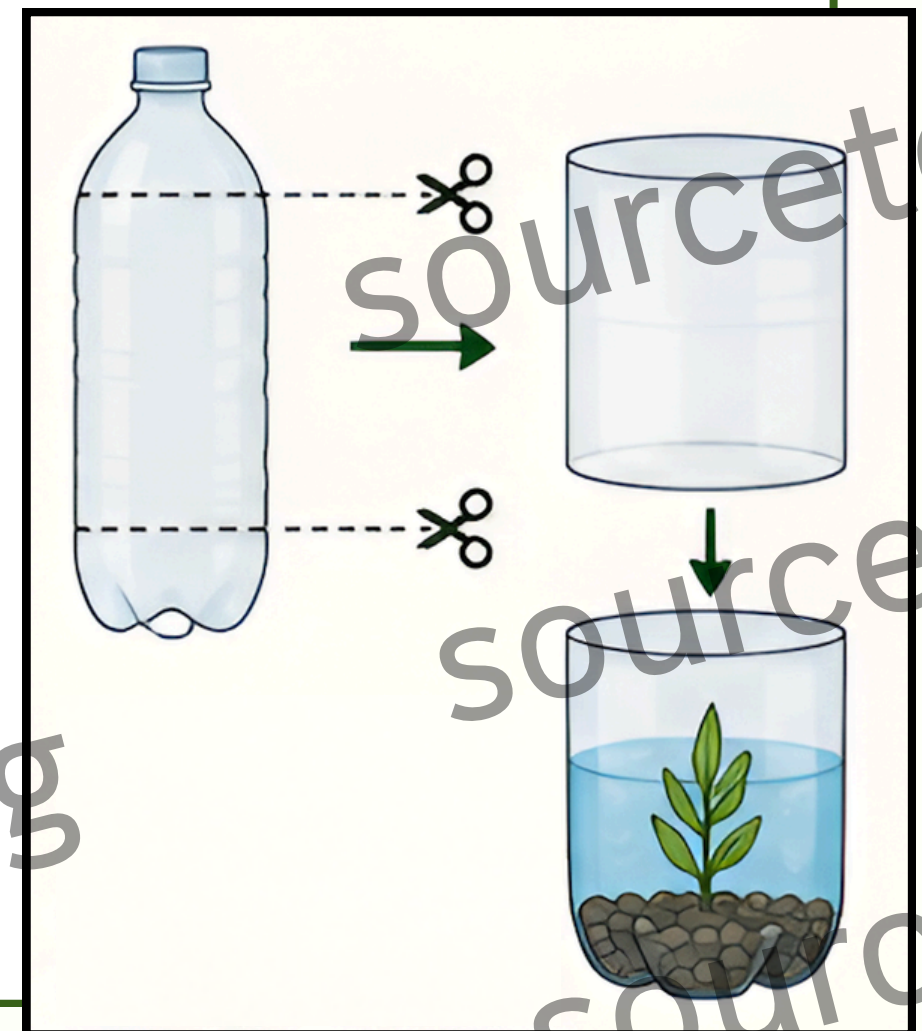
Step 1: Aquatic Chamber

- Cut the first 2-liter bottle in half. Save the top!
- To the bottom half, add gravel to completely cover the bottom, treated water (2/3 full), and 1 aquatic plant.



Step 2: Connector

- Cut the top and bottom off the second bottle.
- Place the middle section of on top of the aquatic chamber and tape them together



BUILDING THE ECOCOLUMN

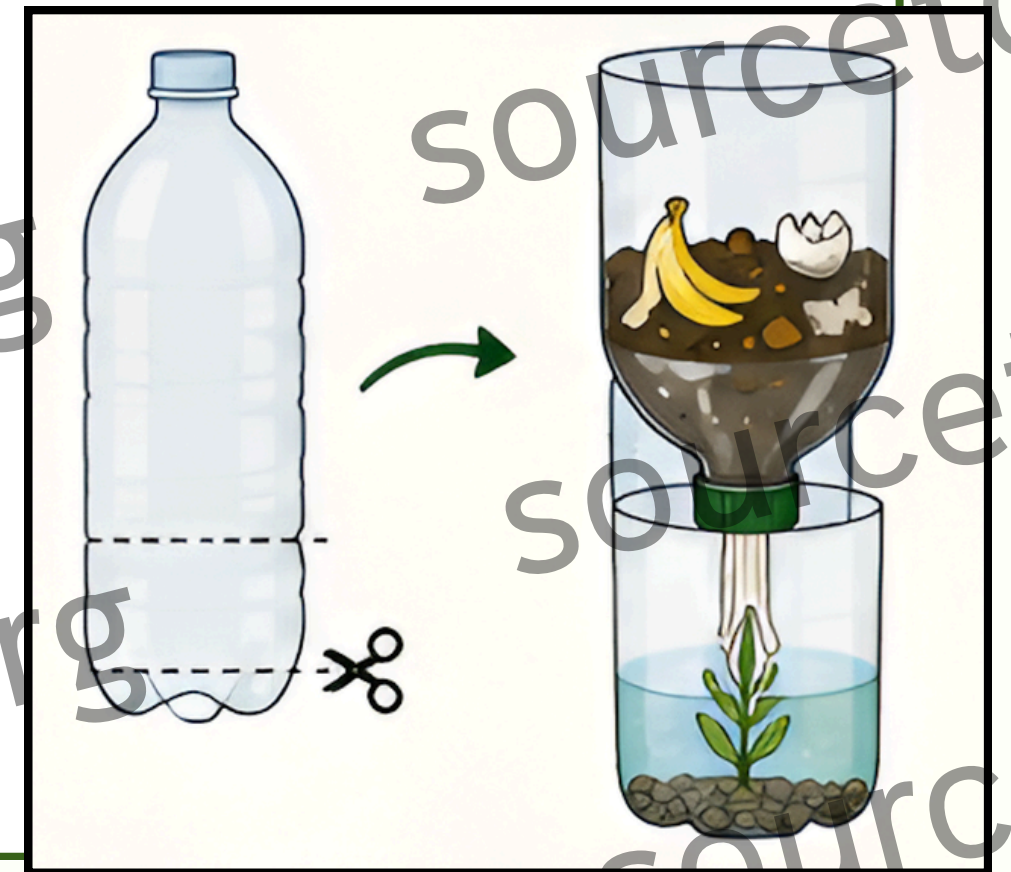
Step 3: Drainage Wick

- Thread cheesecloth through a bottle cap with a hole.
- The wick lets water flow down the chambers and keeps soil in place



Step 4: Decomposition Chamber

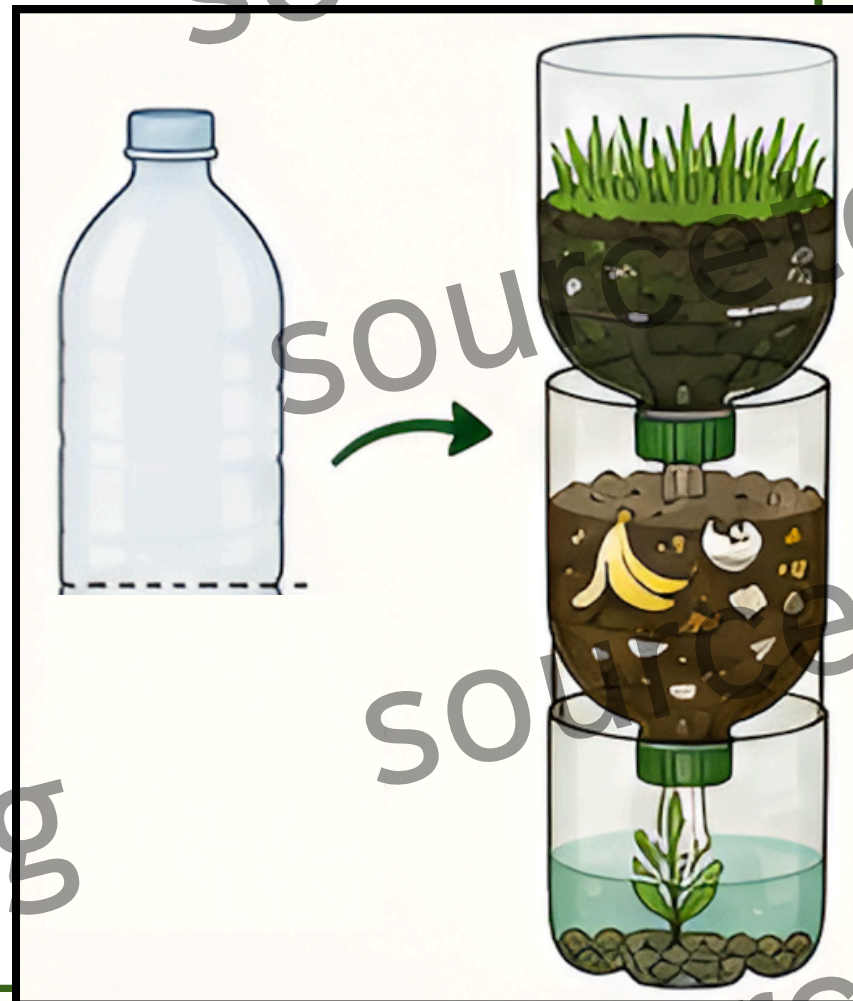
- Cut the top and bottom off the third bottle.
- Screw on the cap with the wick
- Flip the top upside down and place it on top
- Add compost



BUILDING THE ECOCOLUMN

Step 5: Terrestrial Chamber

- Use the top half of the first bottle.
- Repeat Steps 3-4, and place on top of the decomposition chamber.
- Add grass seeds and soil.



Step 6: Top Cover

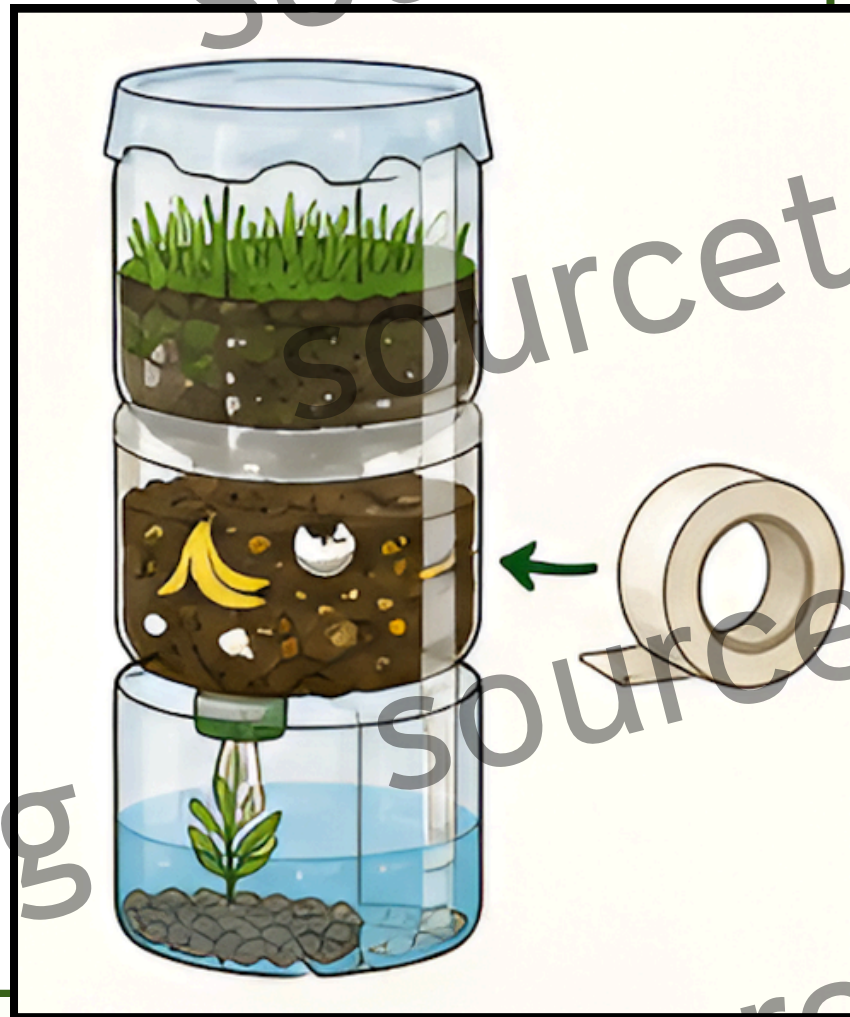
- Place a saved bottle bottom piece over the top as a lid.



BUILDING THE ECOCOLUMN

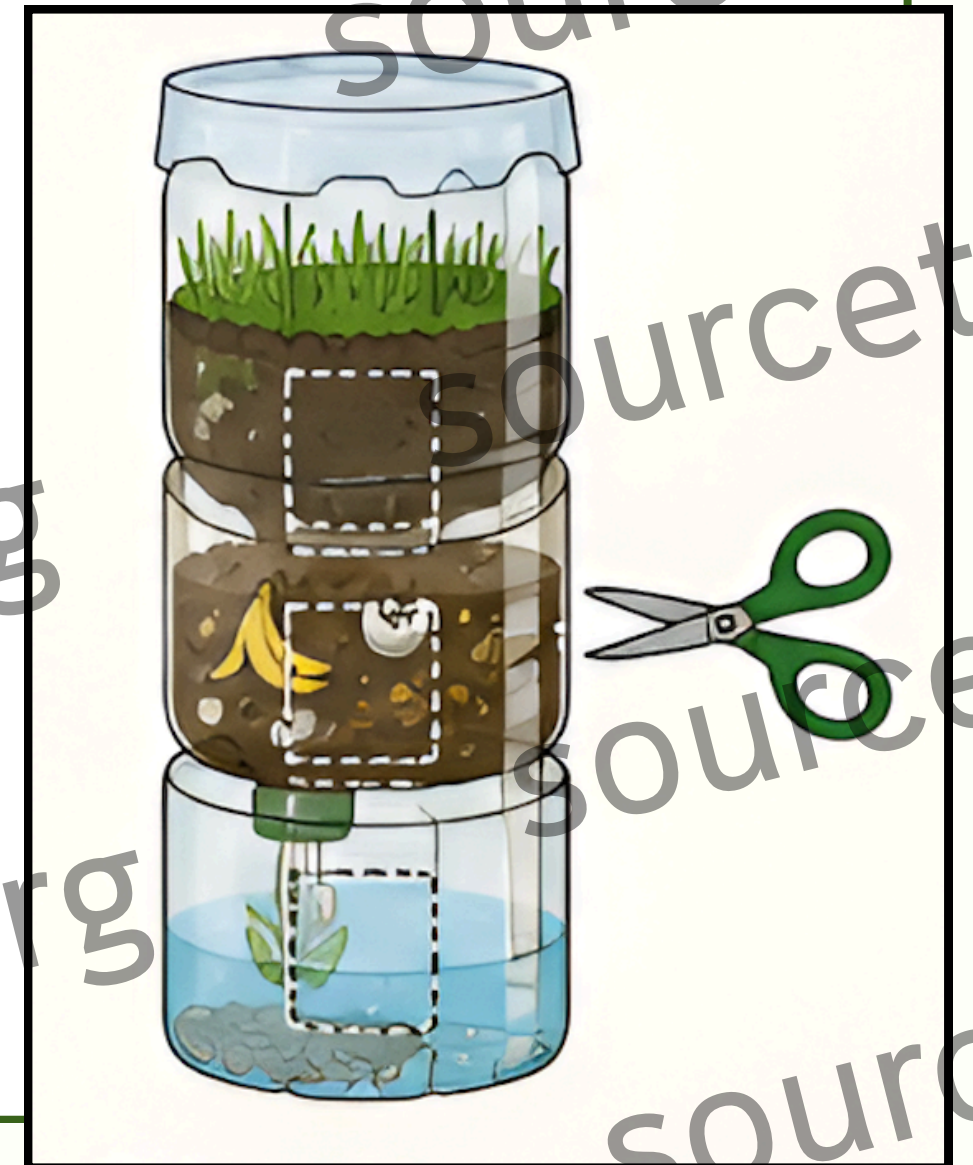
Step 7: Seal It Up!

- Use clear packing tape to seal all connections.
- Make sure there are no leaks!



Step 8: Test Access Flaps

- Cut small square flaps in each chamber for testing access!



TESTING YOUR ECOCOLUMN- SOIL TESTS

Soil tests help us understand the health and fertility of the soil in the terrestrial and decomposition chambers.



IMPORTANCE:

- ✓ Shows how nutrients and acidity change over time
- ✓ Helps us see if our ecosystem is balanced

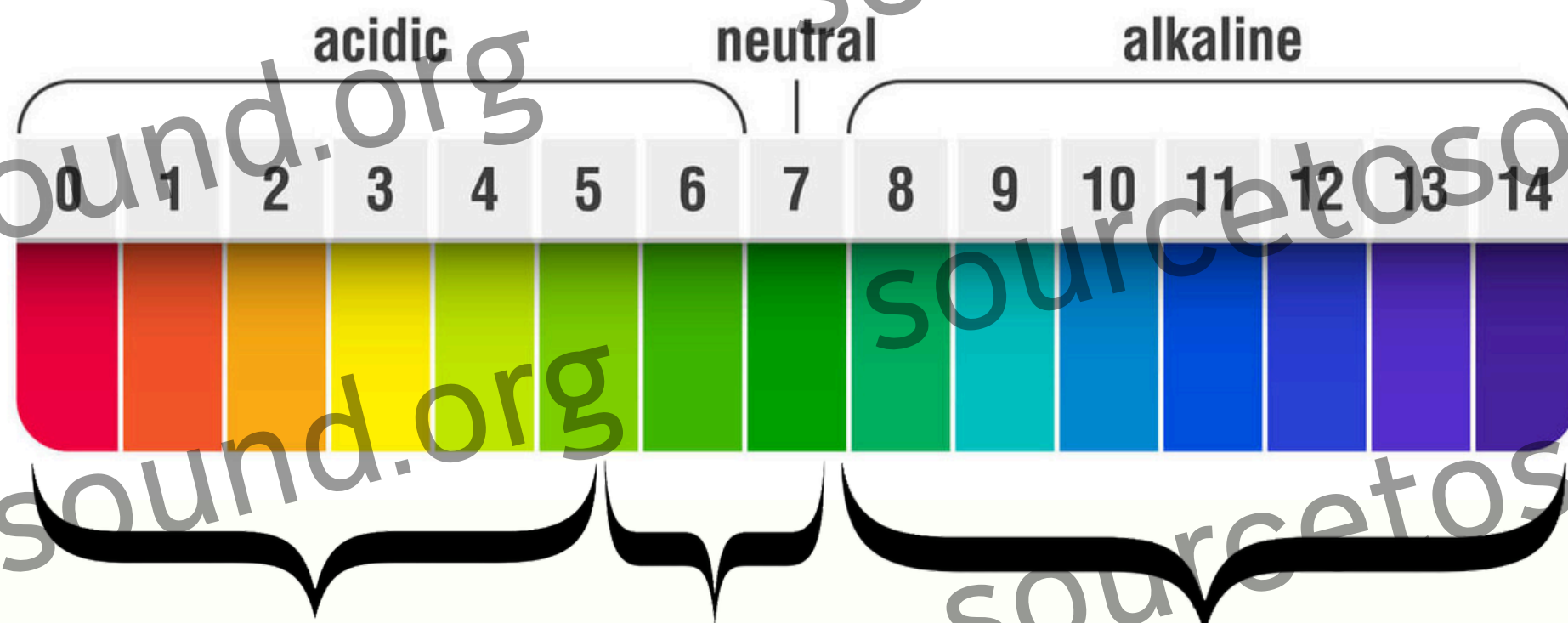
HOW TO PREPARE

- 1 Mix a small scoop of soil with 5 mL of water (1:5 ratio) in a test tube
- 2 Shake tube vertically for 30 seconds
- 3 Place tube in paper cup and wait 10 minutes for soil and water separation

Use separate test tubes for the terrestrial and decomposition chambers!

SOIL PH TEST

pH measures how acidic or basic your soil is.



At pH < 5.5, elements like nitrogen, phosphorous, and calcium are less available for plants to absorption. Also, elements like aluminum and manganese increase to toxic levels.

Ideal soil condition for most plants.

At pH > 7, elements like zinc, copper, iron, and phosphorous become less available in the soil for plants to absorb.

HOW TO PREPARE

- 1 Dip a pH strip into the soil test tube for 3 seconds.
- 2 Hold strip horizontally for color change.
- 3 Compare strip to color chart on test bottle.



SOIL NITRATE/NITRITE TEST

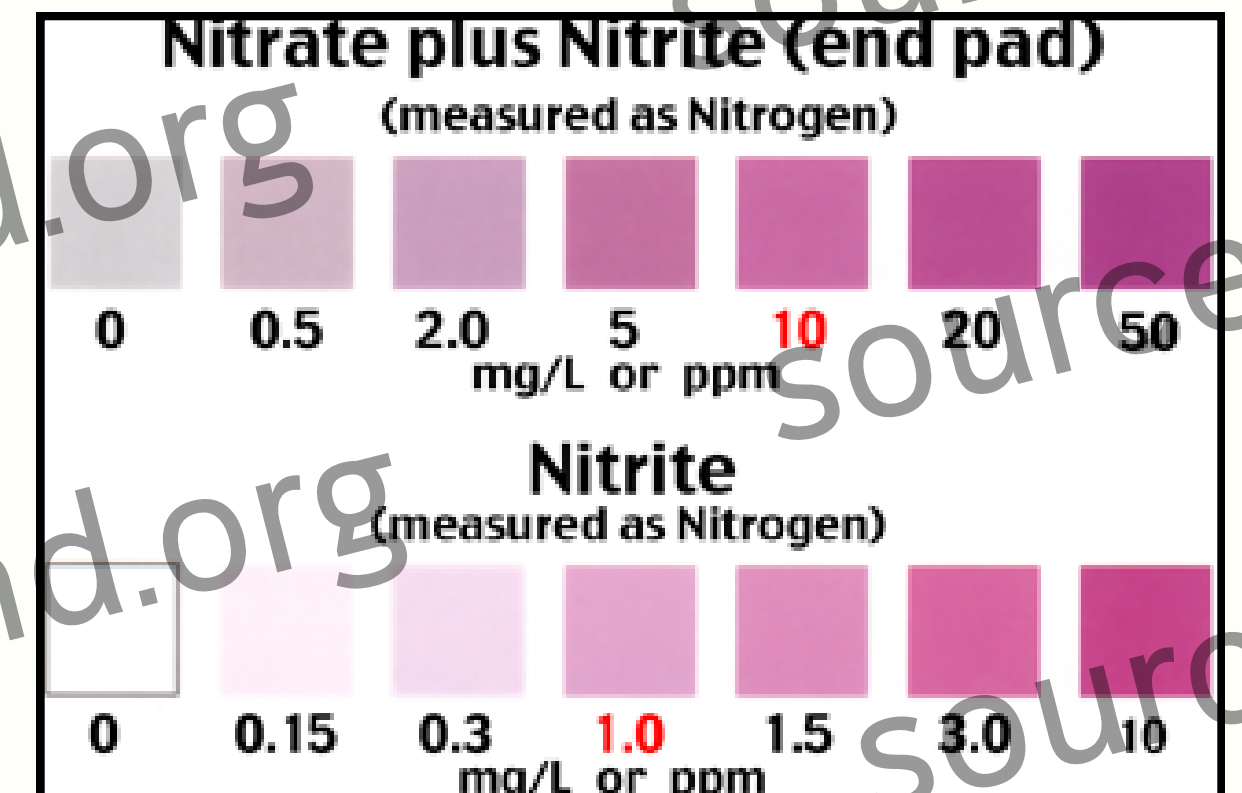
Nitrates and nitrites are nitrogen compounds that build up as organic matter breaks down in the decomposition chamber.

Too much nitrogen draining into the aquatic chamber triggers algae blooms, which consume oxygen and can kill aquatic life.

Tracking nitrate/nitrite levels tells you whether your decomposition chamber is cycling nutrients at a healthy rate, or is overloading the chamber below.

HOW TO PREPARE

- 1 Dip a test strip into the soil test tube for 3-5 seconds.
- 2 Hold strip horizontally for color change.
- 3 Compare strip to color chart on test bottle.



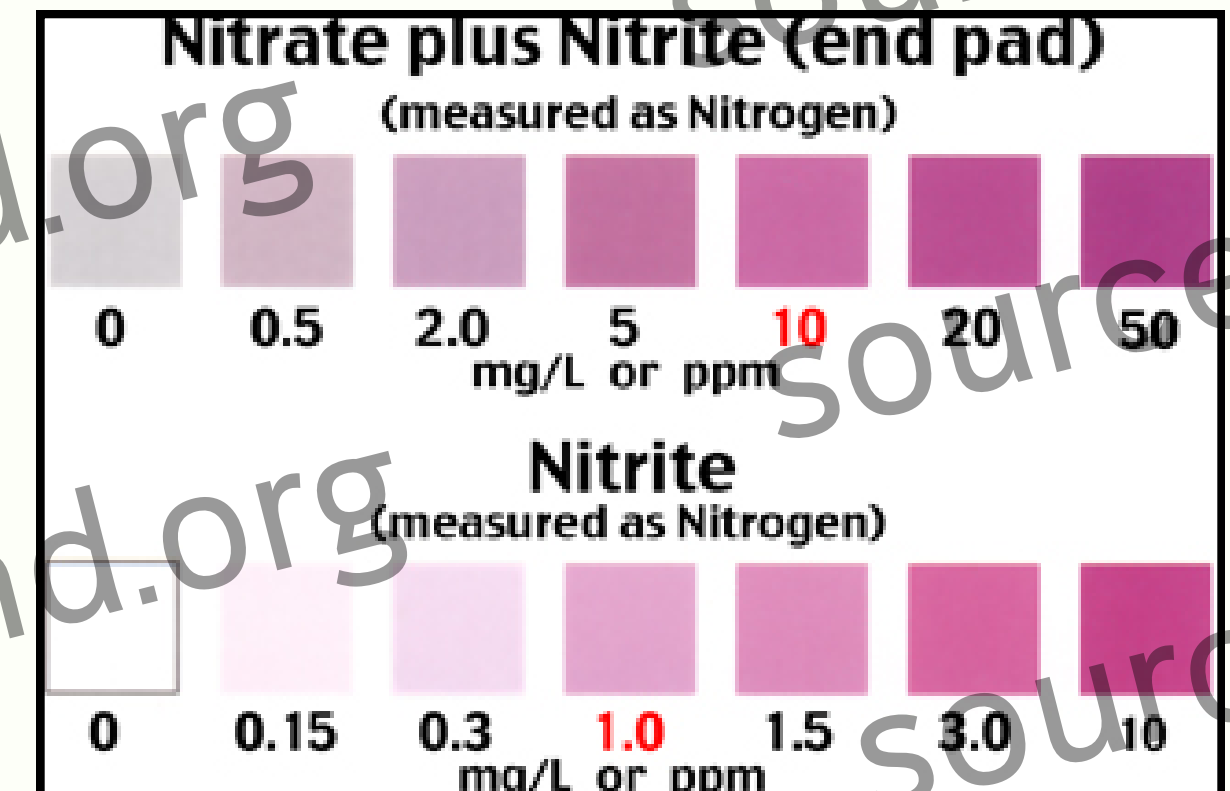
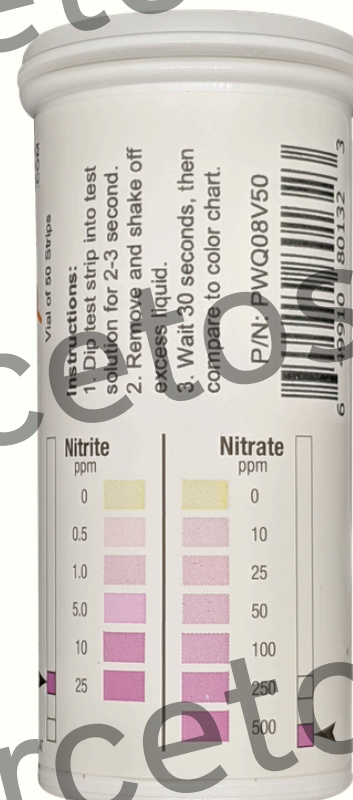
WATER NITRATE/NITRITE TEST

While the soil test measures nitrogen in the decomposition and terrestrial chambers, this test measures nitrates and nitrites in the aquatic water itself.

High levels mean too much nitrogen is draining through to the aquatic chamber, which feeds algae growth and drops dissolved oxygen.

HOW TO PREPARE

- 1 Dip a test strip into the aquatic chamber's water for 3-5 seconds.
- 2 Hold strip horizontally for color change.
- 3 Compare strip to color chart on test bottle.



AQUARIUM 7 IN 1 TEST

The 7-in-1 strip measures general hardness, carbonate, pH, nitrite, nitrate, chlorine, and free chlorine from a single dip into the aquatic chamber.

HOW TO PREPARE

- 1 Dip a test strip into the aquatic chamber's water for 3-5 seconds.
- 2 Hold strip horizontally for color change.
- 3 Compare strip to color chart on test bottle.



Running one strip gives you a broad snapshot of water chemistry, without needing separate tests for each factor.

Changes in multiple readings at once can point to a specific problem in the chambers above.

DISSOLVED OXYGEN (DO) TEST

Dissolved oxygen measures how much oxygen is available in the aquatic chamber's water.

Aquatic plants and organisms need DO about 5 mg/L to survive, so a drop below that signals an issue. Usually, this can be too much algae consuming oxygen during decomposition.

This test is a clear indicator of whether the aquatic chamber is healthy.



HOW TO PREPARE

- 1 Turn on LabQuest and set probe to percentage mode.
- 2 Insert the plug into Channel 1.
- 3 Place the DO probe into the aquatic chamber's water.
- 4 Press play and record for 120 seconds.
- 5 Note down the result.
- 6 Rinse the probe with fresh water, dry it, and screw the protector back on.

COLLECTING DATA

It is important to collect both qualitative and quantitative data to observe and measure changes.

QUALITATIVE

Data collected through observation
with your senses.

EXAMPLE ENTRIES

Date	pH (soil)	Nitrate (ppm)
6/10/2026	7.5	2.0
6/17/2026	7.3	2.2

QUANTITATIVE

Numerical data collected through
measurement and recorded over time.

EXAMPLE ENTRIES

Date	Terrestrial
6/10/2026	The seeds have sprouted a little, with an earthy aroma. There is condensation on the walls.